

1	You are given N bank transactions. Each transaction contains: • Sender • Receiver • Amount • Timestamp A transaction is considered fraudulent if: 1. The Sender, Receiver, and Amount are the same as a previous transaction. 2. The difference between their timestamps is ≤ 60 seconds. Input: 5 ANU JON 200 1000 ANU JON 200 1050 RAM SAM 300 2000 ANU JON 200 2000 RAM SAM 300 2050 Output: ANU JON 200 1050 RAM SAM 300 2050												
2	A gym offers membership plans: <table> <thead> <tr> <th>Months</th> <th>Cost</th> </tr> </thead> <tbody> <tr> <td>12</td> <td>₹15000</td> </tr> <tr> <td>9</td> <td>₹12000</td> </tr> <tr> <td>6</td> <td>₹9000</td> </tr> <tr> <td>3</td> <td>₹5000</td> </tr> <tr> <td>1</td> <td>₹2000</td> </tr> </tbody> </table> Given N months, find the minimum cost to purchase membership. You can buy multiple plans. Input: 15 Output: 20000	Months	Cost	12	₹15000	9	₹12000	6	₹9000	3	₹5000	1	₹2000
Months	Cost												
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3	A hot air balloon has a maximum weight capacity X kg. You are given the weights of N people. Find the maximum number of people that can travel in the balloon such that the total weight does not exceed X. To maximize the count, choose the lightest people first. Input: 6 (People) 4 2 7 1 3 5 10 (Value of X) Output: 4												

4	Given an array of integers (positive, negative, and zero), find the maximum product that can be obtained from any non-empty subset of the array. Input [-1, -1, -2, 4, 3] Output 24 Explanation: $(-1) \times (-1) \times 4 \times 3 = 12$ $(-1) \times (-2) \times 4 \times 3 = 24 \leftarrow \text{Maximum}$
5	A laptop needs a minimum charge X to work. Count how many laptops can work (i.e., charge $\geq$ X). X = minimum charge required for a laptop to work Array A [] containing charge percentages of laptops Input: 30 10 40 50 20 35 Output: 3
6	You are given an integer n representing a number of people initially in happy state In each iteration • 70% happy becomes Sad • 30% of happy remains happy • 50% sad remains Sad • 50% sad becomes Happy Input: 100 3 Output: Happy: 41.2, Sad: 58.8
7	Given an array of N integers (which may contain positive, negative, or zero values), find the maximum possible sum of any contiguous subarray. A subarray is a continuous part of the array. Input: 8 -2 -3 4 -1 -2 1 5 -3 Output: 7
8	Given an array of N integers, find: • The element with the minimum frequency. • The element with the maximum frequency. If multiple elements have the same frequency: • For maximum and minimum frequency, return the element that appears first in the array. Input: 8 1 2 2 3 3 3 4 4 Output: Minimum Frequency Element = 1 Maximum Frequency Element = 3
9	There are N people standing in a circle, numbered from 1 to N. Starting from person 1, every K-th person is eliminated from the circle. After a person is removed, counting continues from the next remaining person. This process continues until only one person remains.

	Your task is to find the number of the last remaining person. Input: 8 4 Output: 6
10	You are given a sorted array of integers of size N and a target value X. (Use Binary Search only) Your task is to find: 1. The index of the first occurrence of X in the array. 2. The index of the last occurrence of X in the array. If X is not present in the array, return -1. Input: 8 1 2 2 2 3 4 4 5 2 Output: 1 and 3
11	You are given a string consisting of lowercase English letters. Encode the string as follows: • For characters at even indices (0-based), shift the character forward by 2 positions in the alphabet (+2). • For characters at odd indices, shift the character backward by 1 position in the alphabet (-1). • Alphabet wrapping is allowed: ◦ 'y' + 2 = 'a' ◦ 'z' + 2 = 'b' ◦ 'a' - 1 = 'z' Return the encoded string. Input: abcdef Output: caecge
12	You are given two integers N and S. Starting from S, find the first prime number greater than or equal to S. Then repeatedly: • Take the next two prime numbers. • Find their sum. • Continue this process until the sum becomes greater than or equal to N. Print the final sum obtained. Input: N = 50 S = 10 Output: 52 Explanation: 11 + 13 = 24 17 + 19 = 36 23 + 29 = 52 > 50 Hence 52 is answer
13	You are given an array of size N containing numbers from 1 to N. One number is missing and one number is repeated exactly once.

	Find the repeating number and the missing number. Input: N = 5 arr = [1, 2, 2, 4, 5] Output: Repeating = 2, Missing = 3
14	Array A is given in ascending order Array B is given in descending order Create a new array of n (size of A) + m (Size of B) Containing all elements from both arrays in ascending order.
15	Alice and her friends are playing a game of verbal Kho-Kho. Alice is acting as a mediator, and the rest of the N friends are seated on N chairs, one each. Alice starts by providing a paper with a single-digit number to the friend present at number 1. Let's denote friends by F, where F will be of size N. F[1]...F[N] represents friends seated respectively. After receiving the paper with a digit, F[1] will enact and try to tell F[2] without speaking. Similarly, F[2] will communicate to the next person i.e., F[3]. This continues until the last person F[N] understands the digit. Finally, the last person will write the digit on a separate paper and give it to Alice to compare both papers. If the digits are similar then, Alice will give a T-shirt to each friend. However, if the digits do not match, Alice will ask each friend's digits, and she will offer the T-shirts to only those who understood the digits correctly. Given N number of friends and digit array D, denoting the digit understood by each friend F. finds out how many of Alice's friends have not enacted well OR did not understand the enactment by the previous friend correctly. Example 1: 3 -> N, number of friends 4 4 4 – array D. denoting digit understanding by N friends Output: 0 Explanation: All of them understood the digits correctly. Example 2: 5 1 2 3 2 2 Output: 4 Explanation: 1st, 2nd, 3rd, and 4th friends could not enact OR understand the enactment.

16	A group of explorers discovers several treasure chests inside an ancient cave. Each chest contains a certain number of gold coins. The chief explorer has a magical lock that can only be opened if the total number of coins in exactly two chests equals a target value K. Your task is to find the indices of the two chests whose coin values add up to K. It is guaranteed that exactly one valid pair exists. You may not use the same chest twice. Return the indices of the two chests in any order. Time Complexity: $O(N)$ Input: 4 2 7 11 15 9 Output: 0 1
17	Given an array of N integers, remove all duplicate elements while keeping only the first occurrence of each element. After removing the duplicates, reverse the resulting array and print it.
18	At the security checkpoint, airport security personnel have seized a number of travellers' belongings. Everything has been thrown into a big box (array). Each product carries a specific level of risk[0,1,2]. The risk severity of the items in this case is represented by an array[] of N integer values. Sorting the elements in the array according to the degrees of danger is the task at hand. Between 0 and 2 are the risk values. Example : Input : 7 -> Value of N [1,0,2,0,1,0,2]-> Element of arr[0] to arr[N-1], while input each element is separated by new line. Output : 0 0 0 1 1 2 2 -> Element after sorting based on risk severity Example 2: input: 10 -> Value of N [2,1,0,2,1,0,0,1,2,0] -> Element of arr[0] to arr[N-1], while input each element is separated by a new line. Output : 0 0 0 0 1 1 1 2 2 2 ->Elements after sorting based on risk severity. Explanation: In the above example, the input is an array of size N consisting of only 0's, 1's and 2s. The output is a sorted array from 0 to 2 based on risk severity.
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Jack is a sports teacher at St. Patrick's School. He makes games not only to make the student fit but also smart. So, he lined up all the N number classes. of students in his class. At each position, he has fixed a board with the Integer number printed on it. Each number is unique and in exactly the range of N. Let us say there are 10 students, then the boards will be printed with numbers from 1 to 10 in a random order given by the sequence A[]. As a rule, all students wear a jersey with their numbers printed on it. So if there are students, each will have a unique jersey number just like a football team. Now, in the beginning, all the students will stand as per the increasing order of their jersey numbers, from left to right. The only difference will be their respective board number which is placed at their respective location. The board location is fixed and cannot be changed. We can consider the arrangement as below. Suppose there are students, and the board is placed in the order of [2 3 1 5 4].

Board — 2, 3, 1, 5, 4

Student's Jersey — 1, 2, 3, 4, 5

Now the game begins.

After every beat of the drum, each student will have to move to that location (index), where his board is pointing to. In the above case student with jersey #1 is standing with board #2, so now he will have to move to location #2. Similarly, all the other students will do.

So after the first beat of the drum, the alignment will be:

Board — 2, 3, 1, 5, 4.

This keeps going on and on until all the students are back to the way they were at the beginning. So, after 6 beats of the drum, all the students will be aligned the same way as before.

Given N and the order of the board of the respective positions, find the number of beats required to bring back the students to their original position.

So, for the above case, the answer is 6

Example 1:

Input:

3 -> Input integer, N

{1, 2, 3} -> Input integer. B[], board alignment.

Output:

1 -> Output

Explanation:

All the students will be standing in board positions;

Board — 1, 2, 3

Student's Jersey -1, 2, 3

After the first beat of the drum:

Jersey #1 will move to index 1.

Jersey #2 will move to index 2.

Jersey #3 will move to index 3.

Hence, they will be back in their position in just 1 beat.

So, the answer is 1.

Example 2:

Input:

5 -> Input integer, N

{2, 3, 1, 5, 4} -> Input integer, B[], board alignment.

Output:

6 -> Output

Explanation:

All the students will be standing as below, with the board positions:

Board — 2, 3, 1, 5, 4

Student's Jersey — 1, 2, 3, 4, 5

After Beat-1 of the drum:

Jersey #1 has moved to index 2.

Jersey #2 has moved to index 3.

Jersey #3 has moved to index 1.

Jersey #4 has moved to index 5.

Jersey #5 has moved to index 4.

Board — 2, 3, 1, 5, 4

Student's Jersey — 3, 1, 2, 5, 4

After Beat-2 of the drum:

Jersey #3 has moved to index 2.

Jersey #1 has moved to index 3.

Jersey #2 has moved to index 1.

Jersey #5 has moved to index 5.

Jersey #4 has moved to index 4.

Board — 2, 3, 1, 5, 4

Student's Jersey — 2, 3, 1, 4, 5

After Beat-3 of the drum:

Board — 2, 3, 1, 5, 4

Student's Jersey — 1, 2, 3, 5, 4

After Beat-4 of the drum:

Board — 2, 3, 1, 5, 4

Student's Jersey — 3, 1, 2, 4, 5

After Beat-5 of the drum:

Board — 2, 3, 1, 5, 4

Student's Jersey — 2, 3, 1, 5, 4

	After Beat-6 of the drum: Board — 2, 3, 1, 5, 4 Student's Jersey — 1, 2, 3, 4, 5 Hence, they will be back in their positions after 6 beats. So, the answer is 6.
20	Bob is going to bet today on horse riding. There are N horses listed in a sequence of 1 to N. The probability of winning each horse is different so the prices for making a bet on the horses are not the same. There is no limit on the number of horses on which he can bet, but he thinks that if he bets on a continuous sequence of horses then he has a better chance to win. Bob will get K units of money if any horse on which he bets will win. But as the award is only K units so he wants to put money less than K. Bob wants to bet on as many horses as he can. As you are his best friend, he reached out to you for help, can you please find the length of the maximum continuous sequence of horses on which Bob can make a bet, and remember he will invest money less than K units. If there is more than one possible combination, Bob will bet randomly on any one of them. Given the number of horses(N), reward money(K), and price of betting on N horses in order. Hint: For each starting index of a horse, its end index in sequences will be equal to or greater than the end index of the previous starting index. Example 1: Input: 10 100 -> N = 10, K=100 30 40 50 20 20 10 90 10 10 10 -> Price to make bet on each horse in order Output: 3 Explanation: There are 10 horses, and the reward money is 100. So, Bob will put money in less than 100. There are two possible o sequences of length three whose total money for betting is less than 100, i.e. [50 20 20] (sum is 90) and [10 10 10] (sum is 30). Bob will choose randomly one sequence from these two. As none of the other sequences with a length greater than 3 will have a price less than 100 so the answer will be 3. Example 2: Input: 10 100 -> N = 10, K=100 10 90 80 20 90 60 40 60 70 75 -> Price to make bet on each horse in order Output: 1

	Explanation: There are no two consecutive horses for which the sum of price is less than 100. So, Bob will choose randomly any one horse. And the max length of the sequence will be 1.
21	This vacation you went to visit the golden house. There are N rooms in this golden house and its owner needs someone to take care of the management of this house. As you have been unemployed for a long time, you are interested in this job. The owner of this house wanted an intelligent manager for this role, so he created one puzzle within that golden house. The person who will be able to solve that puzzle will be the manager of the golden house. The owner of the house kept some number of golden coins in each room. You have to choose two rooms, one from where you will enter and the other one from where you will exit. From any room either you can exit, or you can move to the next room. While visiting any room you will collect all the gold coins, and if you enter any room then you can't skip collecting gold coins from that room, you have to take those coins. The owner wants to have exactly K coins, when you exit the room, he guarantees that there will be at least one possible solution for this puzzle. Given several rooms (N), and several gold coins in N rooms. You have to find room numbers from where you will start and from where you will exit. If there is more than one solution possible, then you have to provide a solution with a smaller starting room number. You can assume that room numbers will start from 1 and end at N. Hint: Find a continuous subsequence whose sum will be exactly equal to K. Example 1: Input: 10 15 -> N =10, K = 15 5 3 7 14 18 1 18 4 8 3 -> Number of gold coins in each room. Output: 1 3 Explanation: There are ten rooms in the house. You want to have the Total sum of gold coins in a continuous sequence of rooms to be 15 There are two solutions to this i.e. [1, 3] and [8, 10] then the solution with a smaller starting room number will be printed hence [1, 3] is output.
22	For all of its products, a supermarket maintains a pricing structure. Each product has a value N printed on it. The price of the item is determined by multiplying the value N, which is read by the scanner, by the sum of all its digits. The goal here is to create software that, given the code of any item N, will compute the product (multiplication) of all the value digits (price). Example 1:

	Input : 5244 -> Value of N Output : 160 -> Price Explanation: From the input above Product of the digits 5,2,4,4 $5*2*4*4= 160$ Hence, output is 160.
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